

Stage 1: direct ABC baseline comparison

Frozen mug pilot: 4 development episodes, 8 test episodes, 12 timestamps each. Equal ridge calibration on development episodes only; endpoint-excluded analysis recalibrates using 11 timestamps per development episode.

Method	Calibrated MAE	Without final sample	Raw completion Spearman
Constant	0.328	0.313	-
DT	0.226	0.233	0.791
S1	0.301	0.323	0.403
S3	0.297	0.321	0.477
S5	0.296	0.321	0.460
Elapsed time	0.196	0.191	0.935
Robometer-4B	0.316	0.306	0.220
DINO matched	0.323	0.310	0.256
DINO separate	0.338	0.315	0.093
DT + S1	0.226	0.245	-
DT + S3	0.225	0.246	-
DT + S5	0.225	0.249	-

Native Robometer progress (no calibration): MAE 0.344, 95% episode-bootstrap interval [0.287, 0.399], VOC Pearson correlation with time 0.220.

Interpretation: DT has lower completion error than Robometer and DINOv2 in this small ABC pilot. Elapsed time remains stronger. This is not a result on Robometer public benchmarks and does not establish general superiority.

Conditioning differs: Robometer receives the dataset task instruction (with the sim prefix removed) and up to eight frames sampled across the causal prefix. PSI uses current/recent images and a supplied visual goal; DINO uses current and goal images. The matched visual goals come from each successful episode final frame and provide hindsight information. Robometer receives no visual goal. The native task instruction is singular (mug), whereas the state-based completion criterion counts both spawned mugs. All methods use the same top camera; each baseline retains its native preprocessing.

Robometer: official RBM model, collator, special-token order and discrete-bin expectation. All backbone/progress/success weights loaded; six unused obsolete similarity-head tensors are ignored, matching the current upstream architecture. No baseline fine-tuning. DINOv2-base uses normalized CLS cosine similarity with its official image processor.

Other ABC outcome audit

Task	Successful endpoints	Incomplete endpoints
sim_load_the_plates_into_the_dish_rack	102	55
sim_put_the_plastic_bottles_in_the_bin	282	5
sim_set_up_chess_pieces_on_the_board	10	7
sim_sweep_away_paper_scraps_from_the_table	0	55

These are full validation-split counts for the four listed tasks, derived from saved qpos and each assembled scene using pinned ABC evaluator code and verified assets. Incomplete means the released evaluator did not report success at the recorded endpoint; it does not establish why execution stopped. The sweep criterion needs further validation because all 55 records were unsuccessful. All 368 mug-validation episodes succeeded; 11 contain regressions. A separate five-per-task screen is exploratory, not a failure-rate estimate.

Plan: stage 1 complete; stage 2 is conditional on reliable outcome measures and independently supplied matched-scene goals. The user approved skipping stage 2 if usable ABC success measures cannot be established. Current bottle/plate/chess counts support keeping it under consideration. Public RBM-EVAL-00D follows stage 2 or its documented skip.

Baseline protocol

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{
  "checkpoint_revisions": {
    "Robometer-4B": {
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    "repo": "robometer/Robometer-4B",
    "revision": "beef63bc914c5c189329d49c6d712d96d632aa34"
  },
  "dinov2-base": {
    "repo": "facebook/dinov2-base",
    "revision": "f9e44c814b77203eaa57a6bdbbd535f21ede1415"
  },
  "Qwen3-VL-4B-Instruct": {
    "repo": "Qwen/Qwen3-VL-4B-Instruct",
    "revision": "ebb281ec70b05090aa6165b016eac8ec08e71b17"
  }
},
"robometer_code_revision": "352d160389daa964788delec933d1925f3a6de4f",
"robometer_protocol": "Official RBM model and collator; eight uniformly sampled causal-prefix frames (five in initial short prefix), top camera, saved checkpoint multi-image progress tokens; instruction only; no visual goal. SDPA bf16 inference.",
"dino_protocol": "DINOv2-base CLS cosine similarity with official image processor; current top image versus matched final-frame or independent goal; fp32 inference.",
"conditioning_caveat": "Robometer has language and longer sampled history; PSI/DINO receive a visual goal, which is hindsight in matched condition. Native-system comparison, not an equal-information architecture ablation.",
"goal_mode": "matched_final_frame"
}

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Metrics and paired confidence intervals

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        "raw_score_spearman": null,
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        "standardization_mean": [],
        "standardization_std": [],
        "coefficients": []
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        "mae_95ci": [
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        "valid_correlation_episodes": 8,
        "raw_score_spearman": 0.7913272497210598,
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        "standardization_mean": [
          -68.64767620712519
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        "standardization_std": [
          51.44173406028894
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        "coefficients": [
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          0.22353910478873532
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      }
    }
  },
}

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"S1": {
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  "mae_95ci": [
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    0.3123603443855311
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  "test_episode_macro_spearman": 0.4031861135364723,
  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4031861135364723,
  "features": [
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  "standardization_mean": [
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  "standardization_std": [
    2.102066231191038
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  "coefficients": [
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    0.18670948565570092
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},
"S3": {
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  "mae_95ci": [
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  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4770141560575599,
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  "standardization_std": [
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  "coefficients": [
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    0.1932362767576606
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"S5": {
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  "mae_95ci": [
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  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.4601199293099806,
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  "standardization_std": [
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    0.1970489289365981
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  "Elapsed time": {
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    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.9353970709613129,
    "features": [
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    "standardization_std": [
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      0.27525928339574546
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    "mae_95ci": [
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    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.21954573670094946,
    "features": [
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    "standardization_std": [
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  "DINO matched": {
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    "mae_95ci": [
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    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.2555432845979947,
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    "standardization std": [
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    "coefficients": [
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      0.10612321349648485
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  "DINO separate": {
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    "mae_95ci": [
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    "test_episode_macro_spearman": 0.09286208310200933,
    "valid_correlation_episodes": 8,
    "raw_score_spearman": 0.09286208310200933,
    "features": [
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    "standardization mean": [
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    "standardization std": [
      0.12742552352698344
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    "coefficients": [
      0.3958333333333333,
      0.11385764587108867
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  },
  "DT + S1": {
    "test_episode_macro_mae": 0.22584291248846752,
    "mae_95ci": [
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    "test_episode_macro_spearman": 0.7620091050265163,
    "valid_correlation_episodes": 8,
    "raw_score_spearman": null,
    "features": [
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      "S1"
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      -7.596841479030748
    ],
    "standardization std": [
      51.44173406028893,
      2.102066231191038
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    "coefficients": [
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      0.1774045368724406,
      0.11639007120522918
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  },
  "DT + S3": {
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    "mae_95ci": [
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    "test_episode_macro_spearman": 0.7698585077666522,

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    "raw_score_spearman": null,
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    "standardization_std": [
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      1.9899154448856193
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    "valid_correlation_episodes": 8,
    "raw_score_spearman": null,
    "features": [
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      "S5"
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    "standardization_std": [
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"native_robometer": {
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  "mae_95ci": [
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  "spearman_against_completion": 0.21954573670094946,
  "VOC_pearson_against_time": 0.22009058615789248
},
"paired_comparisons_against_DT": {
  "Robometer-4B": {
    "mae_minus_DT": 0.09039669004012349,
    "paired_95ci": [
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  }
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"DINO matched": {
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  ]
},
"DINO separate": {
  "mae_minus_DT": 0.11205693089097546,
  "paired_95ci": [
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  ]
}
},
"without_final_sample": {
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      "raw_score_spearman": null,
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      "mae_95ci": [
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      "features": [

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    "S1"
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    0.05918963505437536
  ]
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"S3": {
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  "raw_score_spearman": 0.332645547686473,
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  "standardization_std": [
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    0.08056689230617511
  ]
},
"S5": {
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  "mae_95ci": [
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    0.3355565386442006
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  "valid_correlation_episodes": 8,
  "raw_score_spearman": 0.310887420521725,
  "features": [
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  "standardization_std": [
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    0.0945955866555026
  ]
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"Elapsed time": {
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  "mae_95ci": [
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},

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"test_episode_macro_spearman": 0.9212871866683131,
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  0.23626243480912254
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},
"Robometer-4B": {
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  "mae_95ci": [
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},
"DINO matched": {
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"DINO separate": {
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    "coefficients": [
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      0.005709755144138428
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  },
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    "mae_95ci": [
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    "test_episode_macro_spearman": 0.693244921751889,
    "valid_correlation_episodes": 8,
    "raw_score_spearman": null,
    "features": [
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      "S1"
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      -8.201584935188293
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      49.426629337654774,
      0.6359031186720199
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      0.16953046205776087,
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  },
  "DT + S3": {
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    "raw_score_spearman": null,
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      "S3"
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      -74.72921362790194,
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    ],
  },

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```

    "standardization_std": [
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      0.056653863951307955
    ]
  },
  "DT + S5": {
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    "raw_score_spearman": null,
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    "standardization_std": [
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    "coefficients": [
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      0.16191467938483112,
      0.06384806445417271
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  }
},
"native_robometer": {
  "mae": 0.31584467031908303,
  "mae_95ci": [
    0.2673777188915251,
    0.3628281055431051
  ],
  "spearman_against_completion": 0.2222887403034305,
  "VOC_pearson_against_time": 0.2361584386912496
},
"paired_comparisons_against_DT": {
  "Robometer-4B": {
    "mae_minus_DT": 0.07285726481619129,
    "paired_95ci": [
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      0.08857861912941362
    ]
  },
  "DINO matched": {
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    "paired_95ci": [
      0.06541272747437346,
      0.08832888157286457
    ]
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  "DINO separate": {
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    "paired_95ci": [

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0.06891218932278947,  
0.09385958345163116  
]  
}  
}  
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Checkpoint loading

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{  
  "missing_keys": [],  
  "unexpected_keys": [  
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    "similarity_head.1.bias",  
    "similarity_head.1.weight",  
    "similarity_head.4.bias",  
    "similarity_head.4.weight"  
  ],  
  "mismatched_keys": [],  
  "error_msgs": []  
}
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